

# Global Ratio Monotonicity for a Killed von Mises Bridge

[COMPLETE-THEOREM MANUSCRIPT — SEAL IN PROGRESS; DO NOT  
SUBMIT.]

Pending relay lemmas are marked [SLOT].]

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## Abstract

For  $\beta > 0$  let  $I_m = I_m(\beta)$  denote modified Bessel functions of the first kind, and set  $a_m = I_m^2[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2]$ ,  $b_m = mI_m^4$ ,  $F_A(t) = \sum_{m \geq 1} a_m \sin(mt)$ ,  $F_B(t) = \sum_{m \geq 1} b_m \sin(mt)$ , and  $E = F_A/(2F_B)$ . The global ratio-monotonicity problem for the surface expansion of a two-dimensional  $SU(2)$  lattice gauge observable is: (i)  $F_B > 0$  on  $(0, \pi)$ , and (ii)  $E' < 0$  on  $(0, \pi)$ , for every  $\beta > 0$ . We prove (i) twice, and prove (ii) for  $0 < \beta \leq 3$  via an exact pair-identity skeleton plus an archived interval certificate executed by two independent interval-arithmetic implementations. The structural core is exact:  $E$  is, as an algebraic identity, the mean of  $\cos \psi$  under the midpoint law of a four-step killed von Mises bridge; the generating kernels reduce, via the Neumann addition theorem, to two-dimensional integrals of a *single* Bessel function whose saddle deficit is an exact sum of two squares; and exact saddle cancellations yield the coefficients of the (verified) closed second-order law  $E = \cos(t/2)(1 - c(t)/\beta) + O(\beta^{-2})$ ,  $c(t) = (4 \cos^2(t/4) - 1)/(2 \cos(t/4) \cos(t/2))$ . Three certified negative results (interval arithmetic, two implementations, nested enclosures) kill every monotone full-path coupling, with an exact mechanism at threshold  $\beta |\cos t| = 3/2$ . For the remaining range we give a two-scale closure map with a verified candidate threshold  $\beta_0 = 15$  and, at the  $\pi$ -endpoint, exact identities for the cubic coefficient  $c_3$  (telescoped alternating form, integral form, parity) together with its verified prefactor law. Every claim is labelled exact / certified / verified; the machine-checked lemmas are Lean 4/Mathlib, machine-checked modulo classical Bessel inputs carried as named hypotheses.

## 1 Introduction

This paper consolidates the resolution status of a two-part positivity problem that arises as the last analytic gap in a surface (character) expansion for two-dimensional  $SU(2)$  lattice gauge theory [5, 6, 7]. The problem is stated entirely in terms of modified Bessel functions and trigonometric series, and we treat it as such; no claim of physical consequence beyond the stated expansion is made, and importance is not inherited by thematic proximity.

Throughout,  $\beta > 0$ ,  $I_m = I_m(\beta)$ , and

$$a_m = I_m^2[(m-1)I_{m-1}^2 + (m+1)I_{m+1}^2], \quad b_m = mI_m^4, \quad m \geq 1. \quad (1)$$

Both sequences decay super-exponentially, so

$$F_A(t) = \sum_{m \geq 1} a_m \sin(mt), \quad F_B(t) = \sum_{m \geq 1} b_m \sin(mt), \quad E(t) = \frac{F_A(t)}{2F_B(t)}, \quad (2)$$

are real-analytic on  $\mathbb{R}$ , and the Wronskian  $W = 2(F'_A F_B - F_A F'_B)$  satisfies  $W = 4F_B^2(E)'$  wherever  $F_B \neq 0$  [7].

**Conjecture 1** (global ratio monotonicity). *For every  $\beta > 0$ : (i)  $F_B > 0$  on  $(0, \pi)$ ; (ii)  $E' < 0$  on  $(0, \pi)$ .*

Part (i) is proved below (twice). Part (ii) is proved for  $0 < \beta \leq 3$  (Theorem 21; exact skeleton plus the archived certificate `certify_thmB.py` and its Arb twin), is supported by an exact structural reduction and a verified truth map for  $\beta \geq 15$ , and is otherwise open; Section 9 states the precise two-scale closure map. The conjecture was born inside this programme (it is the global form of an  $\varepsilon$ -window positivity required by the surface expansion); no external pedigree is claimed.

*Convention 2* (tricotomy). Every assertion below carries one of three labels. *Exact*: a proved identity or inequality. *Certified*: a numerical statement rigorously enclosed by interval arithmetic with outward rounding, together with complete analytic bounds for domains and truncations. Precision-stability re-runs and independent implementations are *software audits*, recorded separately with their declared coverage: the certified negatives `certify_time3_negative` and `certify_bridge_matrix` carry a *full twin audit* (nested re-computation plus a complete independent Arb implementation, both transcripts archived); for `certify_thmB` both implementations exist and the Arb transcript is archived — its row is *amber* only because the `mpmath.iv` canonical transcript is still in progress; the  $[6, 15]$  coverage will carry a *planned pre-registered sampled independent audit* (Remark 15). *Verified*: high-precision numerics with stated grids and precision policies, not a proof. No verified statement is used as a hypothesis of an exact one.

## 2 The theorem and the relay

**Theorem 3** (the Surface Theorem; [SLOT: seal completion]). *For every  $\beta > 0$ : (i)  $F_B > 0$  on  $(0, \pi)$ , and (ii)  $E' < 0$  on  $(0, \pi)$ .*

Part (i) is Theorem 18. Part (ii) is proved by the following relay. Rows marked [SLOT] are pending; rows marked *amber* have one archived witness with the second in progress; all other rows are complete with the stated witnesses.

| range                                              | method                  | artifact                    | transcript | state               |
|----------------------------------------------------|-------------------------|-----------------------------|------------|---------------------|
| $(0, \frac{1}{20}]$                                | small- $\beta$ lemma    | Lem. 22                     | —          | complete            |
| $[\frac{1}{20}, 3]$ , all $t$                      | pair skeleton           | <code>certify_thmB</code>   | Arb        | <i>amber</i>        |
| $[3, 6]$ , $t \in [0.6, \pi - \frac{1.5}{\beta}]$  | series intervals        | <code>certify_bulk</code>   | Arb        | <i>amber</i>        |
| $[6, 15]$ , $t \in [0.6, \pi - \frac{1.5}{\beta}]$ | exp-factored integrator | <code>exp_integrator</code> | [SLOT]     | local box: Prop. 14 |
| $[15, \infty)$ , bulk                              | analytic relay          | [SLOT]                      | —          | pending             |
| $t \in (0, 0.6]$ , $\beta \geq 3$                  | sinc + minor criterion  | [SLOT: sinc-cert]           | —          | design ratified     |
| $t \in [\pi - \frac{C}{\beta}, \pi)$               | $c_3$ + radius          | [SLOT: 4 const.]            | —          | A verified          |

Splice boundaries are explicit and overlapping:  $\beta = 3, 6, 15$ ;  $t = 0.6$  and  $t = \pi - C/\beta$ , with  $C_{\text{WIN}} = 1.5$  a candidate pending  $C_{\text{mirror}} \leq 1.5 \leq C_{\text{quad}}$ . The remainder of this manuscript is the completed material, verbatim from the consolidation paper.

*Remark 4* (supersession of the  $e_4$  route). An earlier plan closed the block  $t \in (0, 0.6]$ ,  $\beta \geq 3$  by an  $e_4$ -coefficient lemma with a fixed radius; it is superseded, not silently deleted. A feasibility check

showed the sinc criterion below dominating with ratios from 15 up to  $10^{80}$  across  $\beta \in [3, 15]$  for  $t \leq 0.3$  (*verified*, design-grade), while the apparent infeasibility of  $(0, 0.6]$  belonged to the cancellation of the series tool, which the exp-factored integrator does not suffer. The ratified route has three components: the exact sinc positivity (Lemma 5), cheap one-dimensional-in- $\beta$  minor-criterion certificates on  $(0, t_*(\beta)]$ , and the extension of the integrator strip from  $t = 0.6$  down to  $t_*(\beta) \approx 0.45$  with splice overlap. The slot's pending content is the full minor criterion with its uniform overlap; the  $e_4$  lemma is retired.

**Lemma 5** (exact; sinc positivity). *Let  $1 \leq m < n$  and  $T_{mn}(t) = (m - n) \sin((m + n)t) + (m + n) \sin((n - m)t)$  as in Section 4. Then  $T_{mn}(t) > 0$  for  $0 < t \leq \pi/(m + n)$ .*

*Proof.* Write  $s = m + n$ ,  $p = n - m \geq 1$ , so  $T_{mn}(t) = s \sin(pt) - p \sin(st)$ . Since  $\sin(x)/x$  is strictly decreasing on  $(0, \pi]$  and  $0 < pt < st \leq \pi$ , we get  $\sin(pt)/(pt) > \sin(st)/(st)$ , i.e.  $st \sin(pt) > pt \sin(st)$ ; dividing by  $t > 0$  gives  $s \sin(pt) > p \sin(st)$ .  $\square$

**Lemma 6** (relay criterion for  $\beta \in [6, 15]$ ). *Let  $\langle f \rangle = \iint_{[-\pi, \pi]^2} K f ds d\alpha$  with  $K = I_1(2\beta R)/R > 0$ , and let  $\bar{E}$  be any rational constant. Then  $E' < 0$  at  $(t, \beta)$  if and only if  $\langle D \rangle > 0$  and*

$$W_c := \langle N_t \rangle \langle D \rangle + \langle (N - \bar{E}D) \partial_t \log K \rangle \langle D \rangle - (\langle N \rangle - \bar{E} \langle D \rangle) \langle D \partial_t \log K \rangle < 0$$

(the  $\bar{E}$ -terms cancel identically). *The certificate encloses the five integrals by exp-factored interval cells:  $z = 2\beta R$  is linearized at the exact rational cell center with interval-gradient remainder; the linear exponential integrates exactly; and the two scaled Bessel factors are  $A(z) = e^{-z} I_1(z)/z$  — strictly decreasing, since  $(\log A)' = I_0/I_1 - 1 - 2/z < 0$  follows from the classical lower bound at  $\nu = 0$  (Amos [1]; [2], Thm. 1) and  $\sqrt{1 + z^2} \leq z + 1$  — enclosed by pure endpoint evaluation, and  $B(z) = e^{-z} H_B(z)$ , where*

$$H_B(z) := \frac{I_0(z) - 2I_1(z)/z}{z^2}, \quad H_B(0) = \frac{1}{8}$$

(positive-coefficient entire series; the subscript distinguishes it from the entry-law kernel  $H(\psi)$ ), enclosed unconditionally by  $[e^{-z_{\text{hi}}} H_B(z_{\text{lo}}), e^{-z_{\text{lo}}} H_B(z_{\text{hi}})]$ ; the identity  $K \partial_t \log K = 32\beta^3 c c' (1 - P - Q) H_B(z)$ ,  $c = \cos(t/4)$ , removes  $I_1(z)/z$  from the covariance integrands exactly. The  $(t, \beta)$ -box layer is Proposition 14; the global  $[6, 15]$  coverage remains the pending content of the relay row.

**Lemma 7** (exact; the  $S$ -free relative criterion). *Let  $S = \sin(t/2)$ ,  $E = \langle N \rangle / \langle D \rangle$ , and suppose  $\langle D \rangle > 0$ . Then for all  $t \in (0, \pi)$  and  $\beta > 0$ ,*

$$\frac{E'(t, \beta)}{-\frac{1}{2}S} = \frac{\langle \cos 2s + \cos \alpha \cos s \rangle + 8\beta^3 \iint (N - ED)(1 - P - Q) H_B(z) ds d\alpha}{\langle D \rangle},$$

and consequently

$$E' < -\frac{1}{4}S \iff \mathcal{C} := \langle \cos 2s + \cos \alpha \cos s \rangle + 8\beta^3 \iint (N - ED)(1 - P - Q) H_B - \frac{1}{2} \langle D \rangle > 0.$$

*Proof.* Since  $D_t = 0$ ,  $E' = [\langle N_t \rangle + \iint (N - ED) \partial_t K] / \langle D \rangle$ . The first term factors  $S$  exactly:  $N_t = -\frac{S}{2}(\cos 2s + \cos \alpha \cos s)$ . So does the second:  $\partial_t K = K \partial_t \log K = 32\beta^3 c c' (1 - P - Q) H_B(z)$  with  $c = \cos(t/4)$ , and  $c c' = \cos(t/4) \cdot (-\frac{1}{4} \sin(t/4)) = -\frac{S}{8}$  by the double angle, so  $\partial_t K = -4\beta^3 S (1 - P - Q) H_B$ . Dividing by  $-\frac{1}{2}S$  cancels  $S$  identically and yields the display. For the equivalence: the denominator  $-\frac{1}{2}S$  is negative on  $(0, \pi)$ , so  $E' < -\frac{1}{4}S$  is the *lower* bound  $E'/(-\frac{1}{2}S) > \frac{1}{2}$ , i.e.  $\mathcal{C} > 0$ .  $\square$

**Corollary 8** (exact; the bilinear quotient-free form). *With  $\Phi := \cos 2s + \cos \alpha \cos s$  (the letter  $A$  stays reserved for the scaled Bessel factor  $e^{-z}I_1(z)/z$ ),  $J_N := \iint N(1 - P - Q)H_B ds d\alpha$ ,  $J_D := \iint D(1 - P - Q)H_B ds d\alpha$ , and  $\langle D \rangle > 0$ ,*

$$E' < -\frac{1}{4}\sin(t/2) \iff \hat{\mathcal{C}} := \langle D \rangle \langle \Phi \rangle + 8\beta^3(\langle D \rangle J_N - \langle N \rangle J_D) - \frac{1}{2}\langle D \rangle^2 > 0.$$

*Proof.* Multiply  $\mathcal{C}$  of Lemma 7 by  $\langle D \rangle > 0$  and substitute  $E = \langle N \rangle / \langle D \rangle$ : the quotient disappears by bilinearity, and the inequality direction is preserved.  $\hat{\mathcal{C}}$  is a bilinear combination of five integrals with no quotient anywhere; it is the assembly's sole target.  $\square$

*Remark 9* (why  $\hat{\mathcal{C}}$  is the assembly's object). In  $\hat{\mathcal{C}}$  the variable  $t$  enters only through  $c = \cos(t/4) \in (\frac{\sqrt{2}}{2}, 1)$ : no kinematically forced vanishing factor and no singularity anywhere in the relay range — the common linear zero of margin and correction at  $t \rightarrow 0$  has been cancelled identically. The endpoint degeneracy is thus removed structurally; uniform positivity still requires explicit uniform estimates, which are the assembly's content. The first-order calibration constant is  $a(t) = [\frac{1}{2} + \frac{1}{8c^2}]/c \leq \frac{3\sqrt{2}}{4} \approx 1.061$  (supremum at  $c = \sqrt{2}/2$ ), so the first-order threshold is  $2\sup a \approx 2.12$ ; the target of the analytic relay is  $\hat{\mathcal{C}} > 0$  uniformly for  $\beta \geq \beta_0$  with the remainder included. At the calibration point  $(t, \beta) = (1.5, 8)$  the measured value is  $\mathcal{C}/\langle D \rangle \approx 0.417$  (verified), against the threshold 0. The built-in unit test: the  $1/\beta$  expansion of  $\mathcal{C}/\langle D \rangle + \frac{1}{2}$  must reproduce  $1 - a(t)/\beta$ ; the audited quantity is  $\beta \cdot$  (deviation from first order), which must converge (an  $O(\beta^{-2})$  remainder halves the  $\beta$ -scaled deviation when  $\beta$  doubles, and divides the unscaled deviation by four).

**Lemma 10** (exact; far-region pointwise and kernel bounds). (a)  $\min_{s,\alpha} \Phi = -\frac{9}{8}$  exactly: with  $u = |\cos s|$  the minimum of  $2u^2 - u - 1$  on  $[0, 1]$  is attained at  $u = \frac{1}{4}$  and equals  $-\frac{9}{8}$ . (b)  $|D| \leq 2$  and, by Corollary 25 ( $|E| < 1$ ),  $|N - ED| \leq |N| + |D| \leq (2|C| + 1) + 2 \leq 5$  and  $|1 - P - Q| \leq 1$ . (c) On the deficit region  $\{(s, \alpha) : 2c - R \geq \Delta_0\}$  one has  $z = 2\beta R \leq z_s - 2\beta\Delta_0$  with  $z_s = 4\beta c$ , and since  $A(z) = e^{-z}I_1(z)/z$  is strictly decreasing with  $A(0^+) = \frac{1}{2}$ ,

$$K = 2\beta e^z A(z) \leq \beta e^{z_s} e^{-2\beta\Delta_0} \quad \text{on the region.}$$

Hence the far region's contribution to each of the five integrals of Corollary 8 is bounded by (pointwise constant)  $\times 4\pi^2 \times \beta e^{z_s} e^{-2\beta\Delta_0}$ , exponentially small relative to the saddle mass; the explicit comparison constant against  $\langle D \rangle$  is part of the assembly (Regions I–II).

*Proof.* (a)  $\cos \alpha \cos s \geq -|\cos s|$  with equality attainable, and  $\cos 2s = 2u^2 - 1$ ; minimizing  $g(u) = 2u^2 - u - 1$  gives  $g(\frac{1}{4}) = -\frac{9}{8}$ . (b) Triangle inequalities;  $|N| \leq |C| |\cos 2s| + |\cos \alpha| (|C \cos s| + \sin^2 s) \leq 2|C| + 1$ . (c)  $R \leq 2c - \Delta_0$  gives  $z \leq z_s - 2\beta\Delta_0$ ; write  $K = I_1(z)/R = 2\beta I_1(z)/z = 2\beta e^z A(z)$  and use the monotonicity of  $A$  (Section on the two scaled functions) with  $A(z) \leq A(0^+) = \frac{1}{2}$  uniformly.  $\square$

**Lemma 11** (exact; the two tail families).  $0 < H_B(z) \leq \frac{1}{4}I_1(z)/z = \frac{1}{4}e^z A(z) \leq \frac{1}{8}e^z$  (term by term: the coefficient of  $(z/2)^{2k}$  in  $H_B$  is  $\frac{1}{4} \frac{1}{(k+2)k!(k+1)!} \leq \frac{1}{8k!(k+1)!}$ , the coefficient in  $\frac{1}{4}I_1(z)/z$  being  $\frac{1}{8k!(k+1)!}$ ). Consequently, writing  $\delta X$  for the far-region part of an integral  $X$  over  $\{2c - R \geq \Delta_0\}$ ,

$$|\delta \langle f \rangle| \leq 4\pi^2 |f|_\infty \beta e^{z_s - 2\beta\Delta_0}, \quad |\delta J_f| \leq \frac{\pi^2}{2} |f(1 - P - Q)|_\infty e^{z_s - 2\beta\Delta_0},$$

for the three  $K$ -integrals and the two  $H_B$ -integrals respectively; on the same region  $z \leq z_s - 2\beta\Delta_0$  also bounds  $e^z$  in  $H_B \leq \frac{1}{8}e^z$ .

**Proposition 12** (exact; Region III propagation). *Let  $\hat{\mathcal{C}}_{\text{near}}$  denote  $\hat{\mathcal{C}}$  with every integral restricted to  $\{2c - R < \Delta_0\}$ . Then for every  $\Delta_0 \in (0, 2c)$  and  $\beta \geq 1$ ,*

$$|\hat{\mathcal{C}} - \hat{\mathcal{C}}_{\text{near}}| \leq 576 \pi^4 \beta^4 e^{2z_s} e^{-2\beta\Delta_0}.$$

*Proof.* Global masses:  $\iint K \leq 4\pi^2 \beta e^{z_s} =: M$  and  $\iint H_B \leq \frac{\pi^2}{2} e^{z_s} =: M_H$  (the pointwise bounds  $K \leq \beta e^{z_s}$ ,  $H_B \leq \frac{1}{8} e^{z_s}$  hold on all of  $[-\pi, \pi]^2$  since  $z \leq z_s$ ). With the pointwise constants of Lemma 10 ( $|\Phi|_\infty \leq 2$ ,  $|D|_\infty \leq 2$ ,  $|N|_\infty \leq 3$ ,  $|N(1 - P - Q)|_\infty \leq 3$ ,  $|D(1 - P - Q)|_\infty \leq 2$ ), Lemma 11 gives  $|\delta\langle D \rangle|, |\delta\langle \Phi \rangle| \leq 2a$ ,  $|\delta\langle N \rangle| \leq 3a$ ,  $|\delta J_N| \leq 3b$ ,  $|\delta J_D| \leq 2b$ , where  $a := M e^{-2\beta\Delta_0}$ ,  $b := M_H e^{-2\beta\Delta_0}$  (note  $4\pi^2 \beta e^{z_s - 2\beta\Delta_0} = a$  and  $\frac{\pi^2}{2} e^{z_s - 2\beta\Delta_0} = b$ ). For any product,  $|XY - X_n Y_n| \leq |\delta X| |Y|_{\max} + |X|_{\max} |\delta Y|$  with  $|\langle f \rangle| \leq |f|_\infty M$ ,  $|J_f| \leq |f(1 - P - Q)|_\infty M_H$ . Summing:  $\langle D \rangle \langle \Phi \rangle$  contributes  $\leq 8aM$ ;  $\langle D \rangle J_N$  and  $\langle N \rangle J_D$  contribute  $\leq 12MM_H e^{-2\beta\Delta_0}$  each (using  $aM_H = bM = MM_H e^{-2\beta\Delta_0}$ );  $\frac{1}{2} \langle D \rangle^2$  contributes  $\leq 4aM$ . Total:  $12aM + 192\beta^3 MM_H e^{-2\beta\Delta_0} = (192\pi^4 \beta^2 + 384\pi^4 \beta^4) e^{2z_s} e^{-2\beta\Delta_0} \leq 576\pi^4 \beta^4 e^{2z_s} e^{-2\beta\Delta_0}$  for  $\beta \geq 1$ .  $\square$

*Remark 13* (the choice of  $\Delta_0$  belongs to the assembly). With these deliberately crude (but fully named) constants, at  $\beta = 15$  the bound forces  $\Delta_0 \approx 1.2$  rather than the design value 0.3: the comparison against the saddle-mass lower bound (whose scale is  $e^{2z_s}/\beta^2 \times \text{const}$ ) consumes  $\ln(576\pi^4 \beta^4 \cdot \beta^2 / \text{const}) \approx 30$  of exponent. Since  $\Delta_0 < 2c$  is free, Regions I–II simply cover deficits up to  $\Delta_0$ ; nothing is lost, and the constant hunt stays honest.

**Proposition 14** (certified; local box:  $(t, \beta) \in [1.5, 1.51] \times [8, 8.05]$ ). *For every  $(t, \beta) \in [1.5, 1.51] \times [8, 8.05]$  one has  $\langle D \rangle > 0$  and  $W_c < 0$ ; hence, by Lemma 6,  $E'(t, \beta) < 0$  on this box.*

*Certificate.* Ball-arithmetic machine certificate (python-flint 0.9.0 / Arb). The executed script is identified by SHA-256 prefix **1d888e99** and is archived byte-exact in the repository, together with the self-contained transcript (both file names carry the hash; the transcript’s provenance header records the full hash, library versions, and stage parameters). Three stages, one transcript. *Point* (1.5, 8): 1,951,141 cells,  $W_c/\langle D \rangle^2 \subset [-0.4546, -0.1752]$ . *Stability* (precision 120,  $\Delta z \leq 0.12$ ): 2,500,336 cells, ratio  $\subset [-0.4304, -0.1978]$ , nested inside the point enclosure. *Box*: union of nine parametric sub-boxes ( $\Delta\beta \leq 0.02$ ,  $\Delta t \leq 0.005$ ), each with strictly negative  $W_c$  enclosure and  $\langle D \rangle > 0$  (worst  $W_c$  upper end  $\leq -2.8 \times 10^{21}$ ), 22,986,387 cells in total, 7558 s. All comparisons are strict ball comparisons; the displayed intervals are outward-rounded supersets of the printed enclosures. This proposition validates the mechanism and calibrates the coverage design; it does *not* close the  $[6, 15]$  row, which stays open until the full region  $6 \leq \beta \leq 15$ ,  $0.6 \leq t \leq \pi - 1.5/\beta$  is covered.  $\square$

*Remark 15* (what certifies, and what audits). The mathematical certificate for each sub-box is its *single* rigorous Arb enclosure: outward-rounded ball arithmetic makes one honest enclosure a proof of the strict inequalities, per sub-box. The precision-120 stability stage (with its nesting inside the point enclosure) and the *planned* pre-registered sampled **mpmath.iv** reproduction are *implementation audits*: they test the software that produced the enclosures, not the inequalities themselves, and they are declared as such in the relay table. This fixes the convention for the  $[6, 15]$  coverage: one certificate-grade Arb enclosure per sub-box; stability and sampling audit the pipeline.

**Lemma 16** (safe square roots in the archived run). *In the archived script (SHA-256 **1d888e99**), the square-root guard clamps a negative lower endpoint to 0 and, when the upper endpoint ball straddles 0, returns the dyadic upper bound  $2^{8-p/2}$  at working precision  $p \geq 90$ . This bound is*

valid throughout the run. Indeed, the guarded inputs are true squares:  $R^2 = 4[c_0^2(1-P-Q)+PQ]$  with  $c_0^2 \leq 1$ ,  $P, Q \in [0, 1]$ , and  $z^2 = 4\beta^2 R^2$  with  $\beta \leq 15$ , so  $|R^2| \leq 16$ ,  $|z^2| \leq 14400$ ; the corresponding balls are built by at most twelve correctly rounded operations on factors of unit size, hence  $|R_{\text{ball}}^2| < 16.01$  and  $|z_{\text{ball}}^2| < 14401 < 2^{14}$  (magnitude means  $|\text{mid}| + \text{rad}$ ). The upper-endpoint ball  $u = \text{mid} + \text{rad}$  involves two exact arf-to-ball conversions and one correctly rounded addition, so  $\text{rad}(u) \leq \text{ulp}(2^{14}) = 2^{15-p}$ . If  $u$  straddles 0 then  $|\text{mid}(u)| \leq \text{rad}(u)$ , hence the true upper endpoint is at most  $2\text{rad}(u) \leq 2^{16-p}$  and its square root is at most  $2^{8-p/2}$  — exactly the coded constant. (In the definitive script the branch is replaced by `abs_upper()`, an automatic outer bound with no ad-hoc constant; the present lemma covers the archived run.)

## The two scaled functions

The certificate's precision rests on three one-line facts. (a)  $A(z) = e^{-z}I_1(z)/z$  is strictly decreasing on  $(0, \infty)$ : by the classical lower bound  $I_{\nu+1}(z)/I_\nu(z) \geq z/(\nu+1+\sqrt{(\nu+1)^2+z^2})$ , valid for  $z > 0$  and  $\nu \geq 0$  (Amos [1]; both members of the classical two-sided pair are proved in [2], Theorem 1), taken at  $\nu = 0$ , together with  $\sqrt{1+z^2} \leq z+1$  (square both sides),  $I_1/I_0 > z/(z+2)$ , which is exactly  $(\log A)' = I_0/I_1 - 1 - 2/z < 0$ ; hence  $A$  is enclosed over any  $z$ -interval by its two endpoint values. (b) The kernel-score product collapses algebraically,  $K \partial_t \log K = 32\beta^3 \cos(t/4)(\cos(t/4))'(1-P-Q)H_B(z)$  (the factor  $I_1(z)/z$  cancels between  $K$  and the denominator of the regular score), so the covariance integrands involve only  $B(z) = e^{-z}H_B(z)$ . (c)  $B$  is enclosed *unconditionally* by  $[e^{-z_{\text{hi}}}H_B(z_{\text{lo}}), e^{-z_{\text{lo}}}H_B(z_{\text{hi}})]$ , since  $e^{-z}$  decreases and  $H_B$  (a positive-coefficient entire series) increases; after centering, the  $B$ -terms are subdominant and this crude-but-safe bound suffices. Together with the exact linear-exponential cell integrals, every source of exponential interval growth is either integrated analytically or carried by a monotone one-dimensional profile — the certificate is structure, not brute force.

## 3 The spectral framework

*Convention 17* (normalizations). On  $(0, \pi)$  define the kernel, entry law, and smoothing operator

$$q_1(u, v) = 2e^{\beta \cos u \cos v} \sinh(\beta \sin u \sin v) = 4 \sum_{m \geq 1} I_m \sin(mu) \sin(mv), \quad (3)$$

$$s_1(u) = \frac{\beta}{2} \sin u e^{\beta \cos u} = \sum_{m \geq 1} m I_m \sin(mu), \quad (4)$$

$$(Qf)(v) = \frac{1}{2\pi} \int_0^\pi q_1(v, u) f(u) du, \quad (5)$$

so that  $Q \sin(m \cdot) = I_m \sin(m \cdot)$ . Both closed forms are classical consequences of the generating function  $e^{\beta \cos \theta} = I_0 + 2 \sum_{m \geq 1} I_m \cos(m\theta)$ ; (4) follows by differentiation. All ten verification artifacts in the repository use exactly these normalizations.

Since  $q_1 > 0$  on  $(0, \pi)^2$  and  $s_1 > 0$  on  $(0, \pi)$ , the walk with step kernel  $q_1$  killed on  $\{0, \pi\}$  is well defined;  $H(\psi) = \beta \sin(\psi/2)I_1(2\beta \cos(\psi/2))$  and  $K_2(\varphi) = I_0(2\beta \cos(\varphi/2))$  appear below. Note  $F_B = Q^3 s_1$  and, by the sine orthogonality computation of Lemma 24,  $F_A$  is the companion cosine-weighted integral.

## 4 Theorem A: positivity of $F_B$

**Theorem 18** (exact).  $F_B(t) > 0$  for all  $t \in (0, \pi)$  and all  $\beta > 0$ .



*First proof (kernel positivity).*  $F_B = Q^3 s_1$  by (5) and  $b_m = mI_m^4$ . Each application of  $Q$  integrates a positive function against the strictly positive kernel (3);  $s_1 > 0$  on  $(0, \pi)$ . Hence  $F_B > 0$  pointwise.  $\square$

*Second proof (symmetric-decreasing convolution).* With  $K_2(\varphi) = I_0(2\beta \cos(\varphi/2))$  and  $P_4 = K_2 * K_2$  (circle convolution; Section 7),  $F_B = -\frac{1}{2}P_4'$ . The function  $K_2$  is even and *strictly decreasing* in  $|\varphi|$  on  $[0, \pi]$  ( $I_0$  strictly increasing,  $\cos(\varphi/2)$  strictly decreasing). By the layer-cake representation,  $K_2 = \int_0^\infty \mathbf{1}_{A_\lambda} d\lambda$  with  $A_\lambda$  centered arcs whose half-lengths  $a(\lambda)$  fill  $(0, \pi)$  continuously (strict decrease). For centered arcs,  $(\mathbf{1}_A * \mathbf{1}_B)(t) = |A \cap (t + B)|$  is even and nonincreasing in  $|t|$  on  $[0, \pi]$ , and *strictly decreasing* on  $||a - b|, \min(a + b, 2\pi - a - b)|$ . Hence  $P_4(t) = \iint |A_\lambda \cap (t + A_\mu)| d\lambda d\mu$  is even, nonincreasing on  $[0, \pi]$ , and strictly decreasing on  $(0, \pi)$  because for every  $t \in (0, \pi)$  a positive measure of pairs  $(\lambda, \mu)$  has  $|a(\lambda) - a(\mu)| < t < \min(a + b, 2\pi - a - b)$  (the half-lengths fill  $(0, \pi)$ ). At the derivative level: each overlap  $t \mapsto |A_\lambda \cap (t + A_\mu)|$  is Lipschitz with a.e. derivative  $-1$  in the strict regime and 0 otherwise (on  $[0, \pi]$ ); since  $P_4$  is analytic, dominated convergence gives  $P_4'(t) = \iint \partial_t |A_\lambda \cap (t + A_\mu)| d\lambda d\mu \leq -\mu\{(\lambda, \mu) : \text{strict at } t\} < 0$  pointwise on  $(0, \pi)$ . Thus  $F_B = -\frac{1}{2}P_4' > 0$  on  $(0, \pi)$ .  $\square$

## 5 Theorem B: ratio monotonicity for $0 < \beta \leq 3$

**Lemma 19** (exact; Wronskian identity). *For real arrays with  $\sum m(|a_m| + |b_m|) < \infty$ ,*

$$W := 2(F_A' F_B - F_A F_B') = \sum_{1 \leq m < n} c_{mn} T_{mn}(t), \quad c_{mn} = a_m b_n - a_n b_m, \quad (6)$$

where  $T_{mn}(t) = (m - n) \sin((m + n)t) + (m + n) \sin((n - m)t)$ .

*Proof.* Product-to-sum:  $T_{mn} = 2[m \cos(mt) \sin(nt) - n \sin(mt) \cos(nt)] =: 2G(m, n)$ , antisymmetric with  $G(n, n) = 0$ ; since  $c_{mn}$  is antisymmetric,  $\sum_{m < n} c_{mn} 2G = 2 \sum_{m, n} a_m b_n G(m, n) = 2[F_A' F_B - F_A F_B']$  (first recorded in [7]).  $\square$

The coefficient input is the weighted Turán-type lemma (Appendix A, self-contained):  $\varphi_m = a_m/b_m$  is strictly increasing in  $m \geq 1$  for every  $\beta > 0$ , hence *every minor is negative*:  $c_{mn} < 0$  for  $m < n$  (exact).

**Lemma 20** (exact; the pair identities and bounds). *For  $t \in (0, \pi)$ , with  $p = n - m$ ,  $q = n + m$ :*

$$(a) \quad T_{12} = 4 \sin^3 t; \quad T_{13} = 16 \sin^3 t \cos t; \quad T_{24} = 8 \sin^3(2t), \text{ so } |T_{13}| \leq 16 \sin^3 t \text{ and } |T_{24}| \leq 64 \sin^3 t.$$

$$(b) \quad T_{m, m+1} \geq 0 \text{ for every } m \geq 1.$$

$$(c) \quad |T_{mn}| \leq \frac{\pi^3}{48} pq(q^2 + p^2) \sin^3 t \text{ for all } m < n.$$

*Proof.* (a)  $3 \sin u - \sin 3u = 4 \sin^3 u$  gives  $T_{12}$  (at  $u = t$ ) and  $T_{24}$  (at  $u = 2t$ );  $T_{13} = 4 \sin 2t - 2 \sin 4t = 4 \sin 2t(1 - \cos 2t) = 16 \sin^3 t \cos t$ . (b)  $T_{m, m+1} = (2m + 1) \sin t - \sin((2m + 1)t) \geq 0$  since  $|\sin(qt)| \leq q \sin t$  on  $(0, \pi)$  (induction on  $q$ ). (c)  $T_{mn}$  and its first two derivatives vanish at  $t = 0$  and  $t = \pi$  (the parities of  $m \pm n$  agree), and  $|T_{mn}'''| \leq pq^3 + qp^3 = pq(q^2 + p^2)$ ; hence  $|T_{mn}(t)| \leq \frac{pq(q^2 + p^2)}{6} \min(t, \pi - t)^3$ , and  $\sin t \geq \frac{2}{\pi} \min(t, \pi - t)$  gives the claim.  $\square$

**Theorem 21** (exact modulo the certified computation `certify_thmB.py`). For  $0 < \beta \leq 3$ :  $W < 0$  on  $(0, \pi)$ , hence  $E' < 0$  on  $(0, \pi)$ .

*Proof.* Drop the adjacent pairs  $m \geq 2$  (each contributes  $c_{m,m+1}T_{m,m+1} \leq 0$  by Lemma 20(b) and  $c < 0$ ); keep  $T_{12}, T_{13}, T_{24}$  exactly and majorize every other pair by Lemma 20(c). Pointwise in  $t$ ,

$$\frac{W(t)}{\sin^3 t} \leq \text{Crit}(\beta) := -4|c_{12}| + 16|c_{13}| + 64|c_{24}| + \frac{\pi^3}{48} \sum_{\substack{p \geq 2 \\ (m,n) \neq (1,3), (2,4)}} pq(q^2 + p^2) |c_{mn}|.$$

The certificate `certify_thmB.py` (Section 10) encloses  $\text{Crit}$  in interval arithmetic over adaptive  $\beta$ -boxes covering  $[1/20, 3]$  ( $I_m$  increasing in  $\beta$  gives rigorous boxes; cutoff  $M = 60$  with the tail itself computed in intervals and closed by a derived, per-box-asserted geometric ratio) and certifies  $\text{Crit} < 0$  on all of  $[1/20, 3]$ ; the script then re-certifies *every* box at precision  $+70$  before reporting success (the box count is invocation- and implementation-dependent; the independent Arb implementation `certify_thmB_arb.py` certifies the same coverage with 86 boxes and a full stability pass, transcript archived). For  $0 < \beta \leq 1/20$ , Lemma 22 applies.  $\square$

**Lemma 22** (exact; small  $\beta$ ). For  $0 < \beta \leq 1/20$ :  $\text{Crit}(\beta) < 0$ .

*Proof.* Write  $\kappa = \beta/2 \leq 1/40$ ,  $L_m = \kappa^m/m!$ ,  $U_m = e^*L_m$  with  $e^* = e^{\beta^2/4} \leq e^{1/1600} < 1.001$ ; then  $L_m \leq I_m \leq U_m$  (positive series). *Leading minor:*  $a_2b_1 \geq (I_1^2I_2^2)(I_1^4) \geq L_1^6L_2^2 = \kappa^{10}/4$  and  $a_1b_2 = (2I_1^2I_2^2)(2I_2^4) \leq 4U_1^2U_2^6 = (e^*)^8\kappa^{14}/16$ , so

$$|c_{12}| \geq \frac{\kappa^{10}}{4} \left(1 - \frac{(e^*)^8\kappa^4}{4}\right) \geq \frac{\kappa^{10}}{4} (1 - 10^{-6}).$$

*The three visible positive terms:*  $a_3 \leq I_3^2 \cdot (2I_2^2 + 4I_4^2) \leq (e^*)^4\kappa^{10}/24 \cdot (1 + 10^{-4})$  and  $b_1 \leq (e^*)^4\kappa^4$  give  $16|c_{13}| \leq 16(a_3b_1 + a_1b_3) \leq 0.70\kappa^{14}$ ;  $64|c_{24}| \leq 64(a_4b_2 + a_2b_4) \leq 10^{-3}\kappa^{14}$  (both products have  $\kappa$ -degree  $\geq 22$ ); the largest remaining weighted pair is  $(1, 4)$ :  $\frac{\pi^3}{48} \cdot 510 \cdot (a_4b_1 + a_1b_4) \leq 0.2\kappa^{18} \leq 10^{-6}\kappa^{14}$ . *All other pairs* (those with  $s := m + n \geq 6$ ,  $p \geq 2$ ): using  $a_nb_m + a_mb_n \leq 2mn(e^*)^8\kappa^{4s-2}[(m-1)!m!]^{-2}n!^{-4} + ((n-1)!n!)^{-2}m!^{-4} \leq s^2(e^*)^8\kappa^{4s-2}$  (the factorial bracket is  $\leq 2$  and  $2mn \leq s^2/2$ ),  $pq(q^2 + p^2) \leq 2s^4$ , and at most  $s$  pairs per level,

$$\frac{\pi^3}{48} \sum_{s \geq 6} (\text{level } s) \leq \frac{\pi^3}{48} (e^*)^8 \sum_{s \geq 6} 2s^7 \kappa^{4s-2} \leq 0.66 \kappa^{22} \cdot 6^7 \sum_{j \geq 0} [(7/6)^7 \kappa^4]^j \leq 3.7 \times 10^5 \kappa^{22} \leq 10^{-7} \kappa^{14},$$

where the level ratio  $s \mapsto s + 1$  is at most  $(7/6)^7 \kappa^4 < 10^{-5} < \frac{1}{2}$  (so the geometric closure is valid), and  $\kappa^8 \leq 40^{-8}$  absorbs the constant. Collecting:

$$\text{Crit} \leq -\kappa^{10}(1 - 10^{-6}) + 0.71 \kappa^{14} = -\kappa^{10}(1 - 10^{-6} - 0.71 \kappa^4) < 0,$$

since  $\kappa^4 \leq 40^{-4} < 4 \cdot 10^{-7}$ . (All numbered constants above are crude upper counts; each step loses factors, none gains.)  $\square$

*Remark 23* (verified; the sharp family and saturation). The sharper per-pair bound  $|T_{mn}| \leq \frac{pq(q^2-p^2)}{6} \sin^3 t$  and the adjacent floor  $T_{m,m+1} \geq 2m \sin^3 t$  (note the sign: the floor is a *lower* bound on a positive quantity) are verified far beyond the range used ( $q \leq 151$ ; 13 pairs to  $(10, 30)$ , max ratio 1.0) and would extend Theorem 21 to  $\beta \approx 3.5$ ; their per-pair certification



is a scheduled upgrade. Beyond that the decomposition is *saturated*: with the exact far block (cancellation included) the criterion fails at  $\beta = 6$  by  $+2.2 \times 10^{14}$  while the true  $W/\sin^3 t$  remains negative ( $-6.9 \times 10^{12}$ ) — a verified adversarial finding about the method, and the reason the large- $\beta$  closure uses different machinery (Sections 7–9).

## 6 The bridge structure of $E$ (exact)

Let  $k = q_1 \circ q_1$  denote the two-step kernel,  $k(\psi, t) = 4 \sum_{m \geq 1} I_m^2 \sin(m\psi) \sin(mt)$ , and define

$$\omega_t(d\psi) \propto (Qs_1)(\psi) k(\psi, t) d\psi \quad \text{on } (0, \pi), \quad (7)$$

the law of the time-2 position (midpoint) of the four-step bridge entering from 0 with law  $s_1$  and conditioned to end at  $t$ .

**Lemma 24** (exact).  $\int_0^\pi (Qs_1) k(\cdot, t) d\psi = 2\pi F_B(t)$  and  $\int_0^\pi \cos \psi (Qs_1) k(\cdot, t) d\psi = \pi F_A(t)$ . Hence, as an identity,

$$E(t) = \mathbb{E}_{\omega_t}[\cos \psi] = \frac{F_A(t)}{2F_B(t)}.$$

*Proof.*  $(Qs_1)(\psi) = \sum m I_m^2 \sin(m\psi)$ . By sine orthogonality the first integral picks the diagonal, giving  $2\pi \sum m I_m^4 \sin(mt)$ . In the second,  $\cos \psi \sin(m\psi) \sin(n\psi)$  integrates to  $\frac{\pi}{4}(\delta_{n,m+1} + \delta_{n,m-1})$ , which produces exactly the weights  $(n-1)I_{n-1}^2 + (n+1)I_{n+1}^2$  of  $a_n$ .  $\square$

**Corollary 25** (exact;  $|E| < 1$ ). For all  $t \in (0, \pi)$  and  $\beta > 0$ ,  $|E(t, \beta)| < 1$ . Indeed  $q_1(u, v) = 2e^{\beta \cos u \cos v} \sinh(\beta \sin u \sin v) > 0$  on  $(0, \pi)^2$  and  $s_1 > 0$ , so  $(Qs_1) > 0$  and  $k = q_1 \circ q_1 > 0$  on  $(0, \pi)$ ; hence  $\omega_t$  in (7) is a genuine probability measure and  $E = \mathbb{E}_{\omega_t}[\cos \psi] \in (-1, 1)$ .

**Lemma 26** (exact; entry law).  $\partial_t k(0^+, \psi) = 2H(\psi)$ , where  $H(\psi) = \beta \sin(\psi/2) I_1(2\beta \cos(\psi/2))$ .

*Proof.* Differentiate in  $\psi$  the squared Graf identity  $I_0^2 + 2 \sum_{m \geq 1} I_m^2 \cos(m\psi) = I_0(2\beta \cos(\psi/2))$  [4, §10.23(ii)] and compare with  $\partial_t k(0^+, \psi) = 4 \sum m I_m^2 \sin(m\psi)$ .  $\square$

**Proposition 27** (certified negative; no monotone full-path coupling). At  $\beta = 20$  the time-3 marginal  $\rho_t(u) \propto (Q^2 s_1)(u) q_1(u, t)$  is not stochastically monotone in  $t$ : with  $a = \frac{4802}{10^4}$ ,  $t_1 = \frac{29621}{10^4}$ ,  $t_2 = \frac{30070}{10^4}$ , the tail masses satisfy  $T(t_2, a) < T(t_1, a)$ , with certified enclosure

$$T(t_2, a) - T(t_1, a) \in [-2.5299150980081786690028107, -2.5299150980081786690028074] \times 10^{-10}.$$

Consequently no monotone coupling of full bridge paths exists at  $\beta = 20$ . The certificate uses exact termwise tails ( $J(m, n, a) = \int_a^\pi \sin mu \sin nu du$  in closed form: no quadrature error), positive-series Bessel enclosures with geometric tails, explicit truncation bounds, nested passes  $(M, \text{prec}) = (120, 350) \subset (140, 500)$ , and two independent interval implementations (*mpmath.iv*; *Arb* via *python-flint*).

**Proposition 28** (certified negatives; the weight–kernel matrix). At  $\beta = 20$ , for every entry weight  $Q^{k-1} s_1$  ( $k = 1, 2, 3, 4$ ) the marginal one  $q_1$ -step from the moving endpoint fails stochastic monotonicity, with certified witnesses sharing  $(t_1, t_2) = (3, 61/20)$ :  $k = 1$  (two-step bridge midpoint, the minimal counterexample),  $a = 9/10$ , margin  $\approx -1.34 \times 10^{-2}$ ;  $k = 2$ ,  $a = 3/4$ ,  $\approx$

$-1.80 \times 10^{-6}$ ;  $k = 4$ ,  $a = 3/10$ ,  $\approx -4.75 \times 10^{-13}$  ( $k = 3$  is Proposition 27). Two smoothing steps repair all tested weights (verified); one never does. The statement is relative to the natural weight family:  $k_2$  possesses its own total-positivity corner ( $\beta \gtrsim 8$ ) reachable by two-atom adversarial weights.

**Lemma 29** (exact; the 3/2 mechanism).  $\lim_{u \downarrow 0} \frac{\partial_u \partial_t \log q_1(u, t)}{\beta \sin t u} = 1 + \frac{2\beta}{3} \cos t$ . For  $t > \pi/2$  the corner sign change of  $\partial_u \partial_t \log q_1$  occurs exactly when  $\beta |\cos t| > 3/2$  — the same threshold as the  $TP_2$  failure of  $q_1$ .

*Proof.*  $\partial_t \log q_1 = -\beta \sin t \cos u + \beta \cos t \sin u \coth(\beta \sin t \sin u)$  (direct differentiation of (3)); expand  $\coth z = z^{-1} + z/3 + O(z^3)$  and differentiate in  $u$ ; the stated limit is a one-line computation (symbolic residual zero).  $\square$

*Remark 30* (verified; the mechanism reading). The identification of this local defect as *the* engine of Propositions 27–28 — sine-type entry weights, vanishing linearly at 0 with mass on both sides of the non-monotone region, flip the covariance  $\text{Cov}_{\rho_t}(\mathbf{1}_{u \geq a}, \partial_t \log q_1)$ , while the flat weight does not — is supported by the full weight–kernel matrix and the  $\beta$ -scan of the violations (onset at  $\beta \gtrsim 12$ , consistent with the threshold), but is a verified reading, not a theorem.

## 7 The single-Bessel reduction and the master formula

**Lemma 31** (exact; Neumann product formulas). For  $x, y \geq 0$ , with  $w = \sqrt{x^2 + y^2 + 2xy \cos \alpha}$ ,

$$I_0(x)I_0(y) = \frac{1}{2\pi} \int_0^{2\pi} I_0(w) d\alpha, \quad I_0(x)I_1(y) = \frac{1}{2\pi} \int_0^{2\pi} I_1(w) \frac{y + x \cos \alpha}{w} d\alpha.$$

*Proof.* The first is the integrated Graf/Neumann addition theorem [4, §10.23(ii)]; the second is its  $\partial_y$ -derivative.  $\square$

*Remark 32* (signs). Both formulas extend to all real  $x, y$ , as used below (on  $[-\pi, \pi]^2$  one has  $c_1 = \cos(\varphi/2) \geq 0$  but  $c_2$  may be negative):  $w(x, -y, \alpha) = w(x, y, \pi - \alpha)$ , and the substitution  $\alpha \mapsto \pi - \alpha$  over a full period maps the first identity to itself (both sides even in  $y$ ) and the second to its negative on both sides simultaneously ( $I_1$  odd;  $(-y + x \cos(\pi - \alpha)) = -(y + x \cos \alpha)$ ).

With  $K_2(\varphi) = I_0(2\beta \cos(\varphi/2)) = \sum_{m \in \mathbb{Z}} I_m^2 e^{im\varphi}$  and  $P_4 = K_2 * K_2$  (circle convolution),  $P_4 = \sum_m I_m^4 e^{imt}$  and  $F_B = -\frac{1}{2}P_4'$ . Applying Lemma 31 to both convolution factors:

**Lemma 33** (exact; master representation). Let  $s = \varphi - t/2$ ,  $c_1 = \cos(\varphi/2)$ ,  $c_2 = \cos((t - \varphi)/2)$ ,  $P = \sin^2(s/2)$ ,  $Q = \sin^2(\alpha/2)$ , and

$$R^2 = c_1^2 + c_2^2 + 2c_1c_2 \cos \alpha = 4[\cos^2(t/4)(1 - P - Q) + PQ]. \quad (8)$$

Then

$$F_B(t) = \frac{\beta \sin(t/2)}{8\pi^2} \iint_{[-\pi, \pi]^2} \frac{I_1(2\beta R)}{R} [\cos^2(s/2) - \sin^2(\alpha/2)] ds d\alpha, \quad (9)$$

and, under the common positive kernel  $K = I_1(2\beta R)/R$ ,

$$E(t) = \frac{\langle \cos(t/2) \cos 2s + \cos \alpha [\cos(t/2) \cos s - \sin^2 s] \rangle}{\langle \cos s + \cos \alpha \rangle}, \quad (10)$$

where  $\langle \cdot \rangle$  denotes the  $K ds d\alpha$ -integral over  $[-\pi, \pi]^2$ . The prefactor collapse behind (9) is the sum-to-product symmetrization  $\frac{1}{2}[\sin(\frac{t-\varphi}{2})(c_2 + c_1 \cos \alpha) + (\varphi \leftrightarrow t - \varphi)] = \frac{1}{2} \sin(t/2)(\cos s + \cos \alpha)$ , and  $\cos s + \cos \alpha = 2[\cos^2(s/2) - \sin^2(\alpha/2)]$ .

*Proof.* (i) The squared Graf identity gives  $K_2(\varphi) = \sum_m I_m^2 e^{im\varphi}$ ; convolution multiplies Fourier coefficients, so  $P_4 = \sum_m I_m^4 e^{im\varphi}$  and, termwise (super-exponential decay),  $F_B = -\frac{1}{2}P_4'$ . (ii) Insert Lemma 31 (first formula) with  $x = 2\beta c_1$ ,  $y = 2\beta c_2$  into  $P_4(t) = \frac{1}{2\pi} \int K_2(\varphi) K_2(t - \varphi) d\varphi$ : a two-dimensional integral of  $I_0(2\beta R)$  with  $R$  as in (8); the  $(P, Q)$  form of  $R^2$  is product-to-sum. (iii) Differentiate under the integral (smooth integrand, compact domain):  $\partial_t I_0(2\beta R) = I_1(2\beta R) \cdot 2\beta \partial_t R$  and  $2R \partial_t R = 2(c_2 + c_1 \cos \alpha) \partial_t c_2$  with  $\partial_t c_2 = -\frac{1}{2} \sin((t - \varphi)/2)$ . (iv) Symmetrize over  $\varphi \leftrightarrow t - \varphi$  ( $R$  invariant): with  $\sin(x/2) \cos(x/2) = \frac{1}{2} \sin x$  and the angle-addition formula,

$$\begin{aligned} \frac{1}{2}[\sin(\frac{t-\varphi}{2})(c_2 + c_1 \cos \alpha) + \sin(\frac{\varphi}{2})(c_1 + c_2 \cos \alpha)] &= \frac{1}{4}[\sin(t - \varphi) + \sin \varphi] + \frac{1}{2} \sin(\frac{t}{2}) \cos \alpha \\ &= \frac{1}{2} \sin(\frac{t}{2})(\cos s + \cos \alpha), \end{aligned}$$

using  $\sin(t - \varphi) + \sin \varphi = 2 \sin(t/2) \cos s$  and  $\sin(\frac{t-\varphi}{2})c_1 + \sin(\frac{\varphi}{2})c_2 = \sin(t/2)$ . Collecting constants yields (9). The same substitution in the companion representation of  $F_A$  (Lemma 24 with Lemma 31, second formula) and cancellation of the common factor  $\beta \sin(t/2)$  yields (10). As an end-to-end check, (9) and (10) match the quartic series with ratio 1.000000000000 at  $t \in \{0.4, 1.1, 2.3, 2.9\}$ ,  $\beta = 4$ .  $\square$

**Lemma 34** (exact; geometry of the  $(P, Q)$  square,  $0 < t < \pi$ ). *On  $[0, 1]^2$ : the main saddle is  $(0, 0)$  with  $R = 2 \cos(t/4)$ ; the secondary saddle is  $(1, 1)$  with  $R = 2 \sin(t/4)$  and gap  $2(\cos(t/4) - \sin(t/4)) > 0$  degenerating only at  $t = \pi$ ; the zero set  $\{R = 0\}$  consists of exactly the two corners  $(1, 0)$  and  $(0, 1)$ , and at both of them both brackets of (9) and (10) vanish, so the singular points of  $K$  are switched off (the extension  $I_1(z)/z \rightarrow \frac{1}{2}$  makes  $K$  continuous with value  $\beta$ ). At  $t = 0$  the zero set degenerates to the edges  $P = 1$ ,  $Q = 1$ . The saddle deficit is the exact sum of two squares*

$$4 \cos^2(t/4) - R^2 = 4 \cos^2(t/4) \sin^2(s/2) + 4c_1 c_2 \sin^2(\alpha/2), \quad c_1 c_2 = \cos^2(t/4) - \sin^2(s/2), \quad (11)$$

with sign region  $c_1 c_2 \geq 0 \iff |s| \leq \pi - t/2$ .

*Proof.* (8) and (11) are product-to-sum computations (symbolic zeros).  $R^2 \geq (|c_1| - |c_2|)^2$  with equality forcing  $|c_1| = |c_2|$  (i.e.  $s = 0$  or  $s = \pm\pi$ ) and  $\cos \alpha = \mp \text{sign}(c_1 c_2)$ , which lands exactly on  $(P, Q) \in \{(0, 1), (1, 0)\}$ . Bracket vanishing at both points is direct substitution. The Jacobian of  $(s, \alpha) \mapsto (P, Q)$  carries the arcsine weight  $ds d\alpha = dP dQ / \sqrt{P(1-P)Q(1-Q)}$  (per quadrant), singular at all four corners; near the main saddle one therefore works in  $(s, \alpha)$ , and in  $(P, Q)$  only for the global geometry.  $\square$

**Lemma 35** (exact; saddle cancellations). *Write  $N, D$  for numerator and denominator brackets of (10),  $C = \cos(t/2)$ ,  $c_0 = \cos(t/4)$ . At the saddle  $(s, \alpha) = (0, 0)$ :*

$$\frac{N}{D} \Big|_{(0,0)} = C, \quad N_{\alpha\alpha} - C D_{\alpha\alpha} = 0, \quad N_{ss} - C D_{ss} = -4C - 2,$$

and for  $N_t = -\frac{1}{2} \sin(t/2)[\cos 2s + \cos \alpha \cos s]$ :

$$N_{t,\alpha\alpha} - m D_{\alpha\alpha} = 0 \quad (m = -\frac{1}{2} \sin(t/2)), \quad N_{t,ss} - m D_{ss} = 2 \sin(t/2).$$

Also exact:  $2C + 1 = 4c_0^2 - 1$  and  $C c(t) = 2c_0 - 1/(2c_0)$  for  $c(t) = (4c_0^2 - 1)/(2c_0 C)$ .

**Proposition 36** (verified; the second-order laws). *The frozen-kernel expansion built on Lemma 35 (with  $\langle s^2 \rangle = 1/(\beta c_0) + O(\beta^{-2})$  from (11)) predicts*

$$E = C \left( 1 - \frac{c(t)}{\beta} \right) + O(\beta^{-2}), \quad E' = -\frac{1}{2} \sin(t/2) + \frac{\sin(t/4)(4c_0^2 + 1)}{8\beta c_0^2} + O(\beta^{-2}). \quad (12)$$

*The coefficient identities are exact (Lemma 35); the laws with remainder are verified, not yet certified: measured remainders are  $\approx 0.15/\beta^2$  with true convergence at fixed  $t$  (e.g.  $\beta^2$ -scaled residues 0.1776/0.1767/0.1762 at  $t = 1, \beta = 30/60/120$ ), and  $\beta(E' + \frac{1}{2} \sin(t/2)) \rightarrow \sin(t/4)(4c_0^2 + 1)/(8c_0^2)$  at  $t = 1, 2$ . The remainder control is exactly what the closure map owes. Note the  $1/\beta$  correction to  $E'$  is positive: it opposes the target inequality and must be beaten, not used. The  $\alpha$ -cancellations imply every kernel correction multiplies a vanishing fluctuation and enters at  $O(\beta^{-2})$  only.*

*Remark 37* (exact decomposition of  $E'$ ). With the normalized positive measure  $d\mu \propto K ds d\alpha$ ,  $E' = \langle N_t \rangle / \langle D \rangle + \langle (N - ED) \partial_t \log K \rangle / \langle D \rangle$ ,  $D_t = 0$ , and the regular form  $\partial_t \log K = 4cc'(1 - P - Q)[zI_0(z)/I_1(z) - 2]/R^2$ ,  $z = 2\beta R$ , extends continuously by  $4\beta^2 cc'(1 - P - Q)$  at  $R = 0$ . Verified split at  $(\beta, t) = (5, 1.3)$ :  $-0.238212 - 0.024552 = -0.262764 =$  the finite difference exactly. Bounds for  $E'$  are taken from this exact quotient — never by differentiating an estimate of  $E$ .

## 8 The $\pi$ -endpoint package

**Lemma 38** (exact; telescoping). *With  $a_m$  as in (1):*

$$A_\infty := \sum_{m \geq 1} (-1)^{m+1} m a_m = 0, \quad \sum_{m \geq 1} (-1)^{m+1} m^3 a_m = - \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2.$$

*Hence  $F_A(\pi - \varepsilon) = c_3 \varepsilon^3 + O(\varepsilon^5)$  with*

$$c_3 = \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2. \quad (13)$$

*Proof.* Split  $m^r a_m = m^r(m-1)I_{m-1}^2 I_m^2 + m^r(m+1)I_m^2 I_{m+1}^2$  and shift  $m \rightarrow m+1$  in the first sum: for  $r = 1$  the two weights cancel ( $m(m+1) - m(m+1) = 0$ ); for  $r = 3$  they combine to  $m(m+1)[m^2 - (m+1)^2] = -m(m+1)(2m+1)$ . Absolute convergence justifies the reindexing;  $\sin(m(\pi - \varepsilon)) = (-1)^{m+1} \sin(m\varepsilon)$  expands the sine.  $\square$

**Lemma 39** (exact; integral form and parity). *Let  $g(u) = \frac{1}{2} I_1(2\beta \cos u) = \sum_{m \geq 0} I_m I_{m+1} \cos((2m+1)u)$  (addition theorem). Then*

$$c_3 = \frac{S_3 - S_1}{24}, \quad S_1 = \frac{2}{\pi} \int_0^\pi g\left(\frac{\pi}{2} - u\right) g'(u) du, \quad S_3 = \frac{2}{\pi} \int_0^\pi g'\left(\frac{\pi}{2} - u\right) (-g''(u)) du.$$

*Moreover  $g(-u) = g(u)$  and  $g(\pi - u) = -g(u)$  (parity of  $I_1$ ), hence  $g'(\pi - u) = g'(u)$ ,  $g''(\pi - u) = -g''(u)$ , and both integrands are invariant under  $u \mapsto \pi - u$ : the two halves  $[0, \pi/2]$  and  $[\pi/2, \pi]$  contribute exactly equally, and one may integrate a half range and double. In particular no cancellation between the two phase maxima  $u = \pi/4$  and  $u = 3\pi/4$  is possible — exactly, not asymptotically.*

*Proof.* The expansion of  $g$  is the Graf addition theorem with unit shift. Quarter turn:  $\cos((2m+1)(\frac{\pi}{2}-u)) = (-1)^m \sin((2m+1)u)$ , so  $g(\frac{\pi}{2}-u) = \sum_m (-1)^m I_m I_{m+1} \sin((2m+1)u)$  while  $g'(u) = -\sum_k (2k+1) I_k I_{k+1} \sin((2k+1)u)$  and  $g''(u) = -\sum_k (2k+1)^2 I_k I_{k+1} \cos((2k+1)u)$ , whence  $g'(\frac{\pi}{2}-u) = -\sum_k (-1)^k (2k+1) I_k I_{k+1} \cos((2k+1)u)$ . Orthogonality of  $\{\sin((2m+1)u)\}$  and  $\{\cos((2m+1)u)\}$  on  $[0, \pi]$  (norm  $\pi/2$ ) gives  $S_1 = -\sum_m (-1)^m (2m+1) (I_m I_{m+1})^2$  and  $S_3 = -\sum_m (-1)^m (2m+1)^3 (I_m I_{m+1})^2$ , so

$$\begin{aligned} \frac{S_3 - S_1}{24} &= \sum_{m \geq 0} (-1)^{m+1} \frac{(2m+1)^3 - (2m+1)}{24} (I_m I_{m+1})^2 \\ &= \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2 = c_3, \end{aligned}$$

using  $(2m+1)^3 - (2m+1) = 4m(m+1)(2m+1)$  and the vanishing of the  $m = 0$  term. The parity claims follow from  $I_1(-z) = -I_1(z)$  applied to  $g(\pi-u) = \frac{1}{2} I_1(-2\beta \cos u)$ , and by differentiating the two symmetry relations.  $\square$

**Proposition 40** (verified; prefactor law).  $c_3 > 0$  for  $\beta \in \{1, 5, 20, 40, 80, 120\}$  at cancellation-adjusted precision (the alternating sum cancels at scale  $e^{-(4-2\sqrt{2})\beta}$ ; evaluations use  $\text{dps} \geq 0.6\beta + 50$ ), and

$$c_3 \sim A \beta^{3/2} e^{2\sqrt{2}\beta}, \quad A = \frac{1}{24 \cdot 2^{1/4} \pi^{3/2}} = 0.0062922 \dots,$$

with measured ratios 0.0058646, 0.0060049, 0.0060996, 0.0061475 at  $\beta = 40, 60, 90, 120$  and  $O(1/\beta)$  gaps. The certified large- $\beta$  branch (pending) must produce the signed prefactor with an explicit  $B$ :  $I_{\text{half}} \geq \frac{A}{2} \beta^{3/2} e^{2\sqrt{2}\beta} (1 - B/\beta) - T(\beta)$ ,  $c_3 = 2I_{\text{half}}$ , read  $\beta^*$ , and close  $[15, B_{\text{fin}}]$ ,  $B_{\text{fin}} = \max\{\beta^*, 10C\}$ , by intervals.

## 9 The closure map and the candidate threshold

**Proposition 41** (verified truth map; all statements on the stated finite grids). Let  $g(t, \beta) = E' + \frac{1}{4} \sin(t/2)$ . At  $\beta = 15$ ,  $g < 0$  at every point of a 13-point grid spanning  $[0.05, \pi - 0.09]$  (independently reproduced on a much finer mesh at 110 digits), with thinnest margin  $-0.00573$  at  $t = 0.05$  (on the seal window  $[0.05, \pi - 0.10]$ ) and  $-0.00216$  at  $\pi - 0.09$ ; the crossing sits at  $\pi - t^* = 0.0890502225$ . At  $\beta = 12$  the truth itself fails at  $\pi - 0.10$  ( $g = +0.025$ ); crossings scale as the mirror window:  $\pi - t^* = 0.1155, 0.0891, 0.0644$  at  $\beta = 12, 15, 20$ , matching  $\sqrt{2}/\beta$ . A sweep  $\beta = 15 \dots 60$  keeps the maximum at  $t = 0.05$ , drifting to  $-0.0061$ : strong uniformity evidence, not proof.

**Conjecture 42** (the pending rows of the relay, quantified). Part (ii) of Conjecture 1 holds with the two-scale decomposition: (a)  $0 < \beta \leq 3$  by Theorem 21; (b)  $\beta \in [3, 15]$  by certified evaluation (the holonomic structure of Section 10 makes the compact cheap); (c)  $\beta \geq 15$  by three uniform blocks (windows per the LIVE map of Section 2, superseding earlier drafts): left edge  $(0, 0.6]$  (by the ratified sinc route of Remark 4: the exact sinc positivity of Lemma 5, the certified minor criterion with control of the high-frequency tail  $m+n > \pi/t$ , and the uniform splice overlap with the integrator strip extended down to  $t_*(\beta) \approx 0.45$ ); bulk  $[0.6, \pi - C/\beta]$  (enclosure of  $\langle s^2 \rangle$  and of the covariance of Remark 37, with the five prototype-to-certificate conversions); right edge with moving boundary: assembly to  $\pi - C/\beta$ , then the quadratic regime of  $c_3$  on  $[\pi - C/\beta, \pi]$  with  $C_{\text{mirror}} \leq C \leq C_{\text{quad}}$  (a fixed quadratic window is provably impossible:  $\kappa \approx 0.11\beta$  grows while

$E(\pi - 0.10)$  stays bounded; retention  $E/\kappa d^2$  at  $d = 2/\beta$  is 0.896, 0.892, 0.890 at  $\beta = 20, 40, 80$  — the validity radius scales as  $1/\beta$ ). The named constants still owed:  $B$ ,  $\beta^*$ ,  $C_{\text{quad}}$ ,  $C_{\text{mirror}}$ , and the remainder function  $T(\beta)$ .

The first-order competition is not the obstacle: the two-term law (12) requires only  $\beta > (1 + 1/(4c_0^2))/c_0 \leq 3\sqrt{2}/2 \approx 2.121$ ; everything above that is rigor slack. The candidate  $\beta_0 = 15$  is near truth-optimal for its edge window (Proposition 41), not padded.

## 10 Verification, certificates, and reproducibility

All numerical artifacts live in the repository

<https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>

(directories `scripts/`, `docs/`); the immutable commit hash for the version audited with this manuscript is recorded in the repository’s hash registry (`docs/SURFACE-CLOSURE-NOTES.md`) and in the arXiv comment field on submission. Each certificate embeds its pinned statement and self-verifies on every execution.

*Certificates* (*interval arithmetic*, *outward rounding*, *nested or stability passes*). `certify_time3_negative.py` and its Arb twin (Proposition 27); `certify_bridge_matrix.py` and twin (Proposition 28); `certify_thmB.py` (Theorem 21: adaptive  $\beta$ -boxes covering  $[1/20, 3]$ ; tails computed in intervals and closed at the level of *complete shells*: each pair maps into the next shell with term ratio  $r$  derived from the termwise-exact  $I_{n+1}(x) \leq xI_n(x)/(2(n+1))$ , and the one new pair per shell is bounded by a double index shift, giving shell ratio  $2r$ , asserted  $< \frac{1}{2}$  per box before the geometric closure; a built-in full stability pass at precision  $+70$ ; the Arb twin `certify_thmB_arb.py` independently certifies the full coverage — 86 boxes, both passes, transcript archived; the `mpmath.iv` canonical transcript is committed on completion of its run, as recorded in the repository hash registry). Bessel enclosures use the positive series with an exact-integer stopping rule and geometric tails; floats appear only in loop-termination heuristics. Nested enclosures prevent silent numerical or truncation degradation; statement auditing and the independent Arb implementation guard against common-mode errors.

*Verified computations*. The master-formula and reduction identities (ratios 1.0 to 12 digits); the truth map of Proposition 41 (dps 80, confirmed independently at 110 digits); the  $c_3$  ladder (Proposition 40); the term-split coefficients of (12) at  $\beta = 60, 120, 240$ .

*Precision policies* (each traceable to a caught artifact):  $\text{dps} \geq 2.2\beta + 40$  for any evaluation with  $t \geq 2$ ,  $\beta > 30$  (mirror cancellation);  $\text{dps} \geq 0.6\beta + 50$  for  $c_3$ -type alternating sums; series cutoffs scale with the maximal argument, and scaled-asymptotic evaluation of  $e^{-z}I_n(z)$  is the default for  $z > 35$ ; every measured coefficient carries a  $\beta$ -scaling sanity check; whatever can be computed from a positive integral representation is.

*Audit trail*. Fifteen numerical or formulation artifacts (“ghosts”) were caught during fabrication and writing by the three-way review protocol; Appendix B tabulates them. We consider the table part of the result: it is the reproducibility record that makes the certificates credible.

*Machine-checked lemmas*. The unit-step and Turán-type coefficient inequalities behind Theorem 21 are formalized in Lean 4/Mathlib (`PhiMonotone.lean`, `FHBesselAmos.lean`, `WronskianIdentity.lean`; axiom oracle [`propext`, `Classical.choice`, `Quot.sound`], no sorry). Mathlib has no Bessel functions: what is machine-checked is the ordered-field implication, with the analytic inputs (three-term recurrence [4, 10.29.1]; the Amos-type bound of



[1, 2, 3], cited verbatim, shift  $\nu \rightarrow \nu + 1$  as in Appendix A) carried as named hypotheses. Correct claim everywhere: *machine-checked modulo classical Bessel inputs carried as named hypotheses*.

## Acknowledgements and provenance

The conjecture, the bridge reading, the reduction, and all certificates were produced inside an audit-first programme in which three independent reviewing voices verify every claim before acceptance; scores, objections, and the complete ghost ledger are archived with the repository history. Nothing in this paper rests on trust in any single computation: every load-bearing negative has two witnesses, and every identity marked exact has a symbolic-zero transcript.

## A The weighted Turán-type lemma, self-contained

**Theorem 43** (exact). *For every  $x > 0$  and every real  $m \geq 1$ , with  $\varphi_m = a_m/b_m$  built from (1) at argument  $x$ :  $\varphi_m < \varphi_{m+1}$ . Equivalently, with  $\rho_\mu = I_{\mu+1}/I_\mu$ ,*

$$\frac{1}{m} \left( \frac{m-1}{\rho_{m-1}^2} + (m+1)\rho_m^2 \right) < \frac{1}{m+1} \left( \frac{m}{\rho_m^2} + (m+2)\rho_{m+1}^2 \right).$$

*Proof.* Two classical inputs, valid for all real orders: the three-term recurrence  $\rho_{m-1}^{-1} = \rho_m + 2m/x$ ,  $\rho_{m+1} = \rho_m^{-1} - 2(m+1)/x$  [4, 10.29.1], and the Amos-type bound

$$\rho_m(x) < U_m(x) := \frac{x}{a + \sqrt{a^2 + x^2}}, \quad a = m + \frac{1}{2},$$

proved best of its form for all real orders  $\geq 0$  by Segura [2]; we use [3, Thm. 2], which bounds  $I_\nu(x)/I_{\nu-1}(x)$  for  $\nu \geq \frac{1}{2}$  by  $x/(\nu - \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2})$ ; setting  $\nu = m+1$  bounds  $\rho_m = I_{m+1}/I_m$  by  $x/(m + \frac{1}{2} + \sqrt{(m + \frac{3}{2})^2 + x^2}) \leq x/(a + \sqrt{a^2 + x^2})$  with  $a = m + \frac{1}{2}$ , which is the displayed bound (the last step uses  $\sqrt{(m + \frac{3}{2})^2 + x^2} \geq \sqrt{a^2 + x^2}$ ). Substitute the recurrences, write  $u = \rho_m \in (0, 1)$ ,  $c = 1/x$ ,  $S = u + 1/u$ ,  $P = 1/u - u$ ; clearing the positive denominators, the claim becomes  $\Delta_m > 0$  where the *exact factorization*

$$\frac{\Delta_m}{2m(m+1)} = (S - 3c)(P - (2m+1)c) + (2m+1)c^2$$

holds as an algebraic identity (expand and collect; the coefficients of  $u^{\pm 2}$  are  $\pm 2m(m+1)$ , the linear-in- $c$  terms are  $-4m(m+1)c[(m-1)u + (m+2)/u]$ , and  $(m-1)u + (m+2)/u = \frac{1}{2}[(2m+1)S + 3P]$ ,  $1/u^2 - u^2 = PS$ ). The second factor is positive because  $U_m$  is the positive root of  $1/w - w = (2m+1)/x$  (the calibration  $1/U_m - U_m = 2a/x$ ) and  $w \mapsto 1/w - w$  is strictly decreasing, so  $\rho_m < U_m$  gives  $P > (2m+1)c$ . The first factor is positive because  $\sqrt{a^2 + x^2} > a$  gives  $U_m < x/(2a)$ , so  $u < x/(2m+1) \leq x/3$  and  $S > 1/u > (2m+1)/x \geq 3c$ . The third term is positive. Strictness is strict throughout. (This is the proof of [6], reproduced so that Theorem 21 is self-contained; the ordered-field implication is machine-checked in `PhiMonotone.lean`, modulo the two classical inputs carried as named hypotheses.)  $\square$

## B Audit table (the ghost ledger)

Each row: what was wrong, how it was caught, and the rule it produced. All entries are documented with transcripts in `docs/SURFACE-CLOSURE-NOTES.md`.

| #  | artifact                                           | catch / rule produced                                                                  |
|----|----------------------------------------------------|----------------------------------------------------------------------------------------|
| 1  | random $TP_2$ scan missed adversarial corners      | targeted corner search; adversarial grids mandatory                                    |
| 2  | verification cell with unbound parameter           | explicit binding rule for inline prints                                                |
| 3  | near- $\pi$ scan at low precision, false violation | precision rule $\text{dps} \geq 2.2\beta + 40$ for $t \geq 2, \beta > 30$              |
| 4  | stale-PDF claim misattributed between voices       | provenance ledger: attribute to exact source                                           |
| 5  | float64 CDF increments $\pm 11$ (impossible)       | positive-representation preference                                                     |
| 6  | $t=2, \beta=240$ residue 0.363 vs 0.317            | boundary $t \geq 2$ included in the dps rule                                           |
| 7  | $I_1$ series truncated at 60 terms, $z \leq 480$   | cutoffs scale with argument; scaled asymptotics for $z > 35$ ; $\beta$ -scaling checks |
| 8  | $c_3$ sign flip at $\beta = 120$ (cancellation)    | alternating-sum rule $\text{dps} \geq 0.6\beta + 50$                                   |
| 9  | $R = 0$ corner misplaced at $(1, 1)$               | complete critical-point census required in prose too                                   |
| 10 | one-saddle prefactor off by exactly 2              | census rule; the factor 2 later proved an exact parity                                 |
| 11 | $I_1$ lower bound dropped the negative tail        | sign audits are a separate mandatory pass                                              |
| 12 | fixed quadratic window at $\pi$ was provably false | interrogate obligations before attempting them                                         |
| 13 | Thm. A second proof: rearrangement misquoted       | layer-cake arc proof substituted (this revision)                                       |
| 14 | Lemma sign error: floor stated as upper bound      | corrected; per-pair statements now carry signs explicitly                              |
| 15 | “exact” labels on verified inequalities            | tricotomy re-audit of every label (this revision)                                      |

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